Independent University, Bangladesh

CSE313 Compiler Construction · Practice Set A · Mid + Quiz · Time: 1h30m

New questions in the Aut-23 shape. Solve on paper first, then check.

Question paper

Mid Question (Total marks: 20)

1. Prove whether the following grammar is Ambiguous or NOT. If the grammar is ambiguous, convert it to unambiguous form. **[7]**

```
B → B or B | B and B | not B | ( B ) | id
```

2. Write the First() and Follow() of the following Grammar. Draw LL(1) Parsing Table and parse the string `[ ] d a b c b` showing all steps. If parsing is not possible, point out the reasons. (b) Parse `a c b b`. V = {E, F, G, H}, T = {a, b, c, d, [,], ε}, S = {E} **[8]**

```
E → a F G | H E
F → b | ε
G → c F | ε
H → d | [ F ]
```

3. Find True/False: **[5]**

(a) Every LL(1) grammar is unambiguous.
(b) The FOLLOW set of a non-terminal can contain ε.
(c) An LL(1) parsing table has one row for each terminal.
(d) Syntax analysis comes after lexical analysis.
(e) A Bottom-Up Parser builds the parse tree from the root down to the leaves.

Quiz Question (Total marks: 10)

1. Convert the following Non-Deterministic Grammar (NDG) to Deterministic Grammar (DG): **[5]**

```
S → i E t S | i E t S e S | a
E → b c | b d | b
```

2. Convert the following Left Recursive Grammar (LRG) to Right Recursive Grammar (RRG): **[5]**

```
S → S a B | S b | c
B → B d | e
```